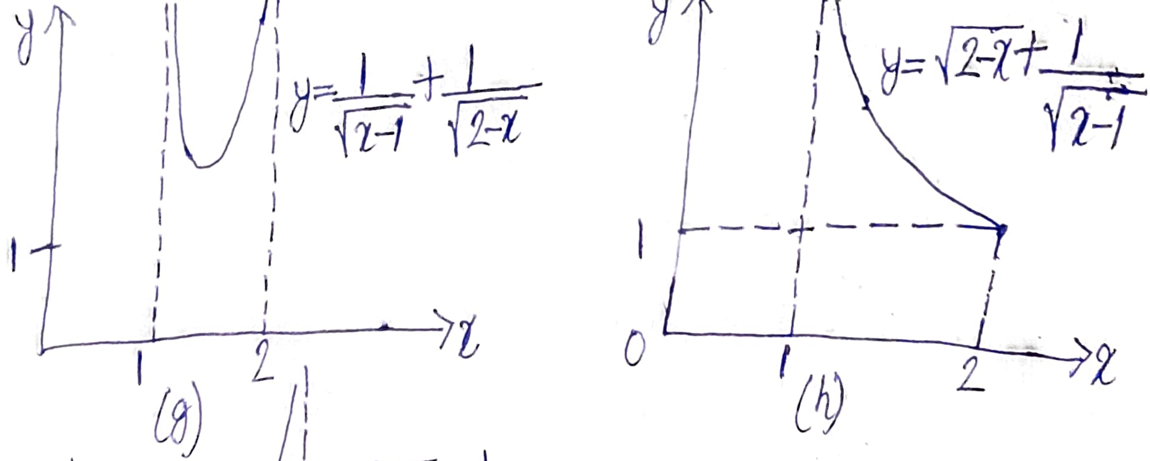

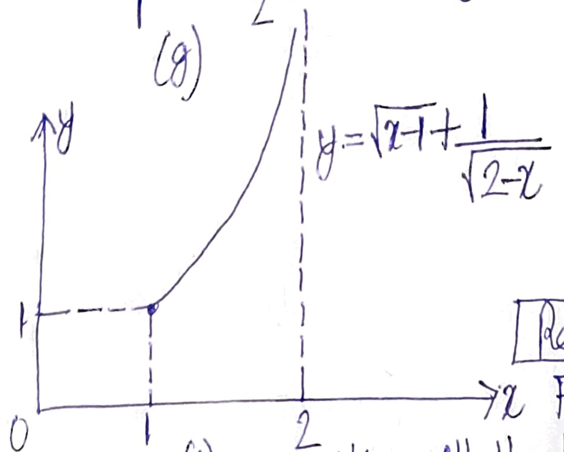

Figure 15

Reader: Obviously, in all the cases shown in Fig. 15 the domain of the function is supposed coinciding with the domain of the corresponding analytical expression.

Author: Yes, you are right. In cases (b), (c), (d) and (e) these domains are infinite intervals. Consequently, only a part of each graph could be shown.

Reader: In other cases, however, such as (g), (h) and (i), the domains of the functions are intervals of finite length. But here as well the figure has space for only a part of each graph.

Author: That is right. The graph is presented in its complete form only in cases (a) and (f). Nevertheless, the behaviour of the graphs is quite clear for all the functions in Fig. 15.

The cases which you noted, i.e. (g), (h), and (i), are very interesting. Here we deal with the unbounded functions defined over the finite interval.

Definition: A function $y=f(x)$ is called bounded over an interval D if one can indicate two numbers A and B such that $\boxed{A \leq f(x) \leq B}$ for all $x \in D$. If not, the function is called unbounded.

Note that within infinite intervals, you may define both bounded and unbounded functions. You are familiar with examples of bounded functions: $y=\sin x$ and $y=\cos x$. Examples of unbounded functions are in Fig. 15 (cases (b), (c), (d) and (e)).